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Lennard Wharton; L. Peter Gold; William Klemperer

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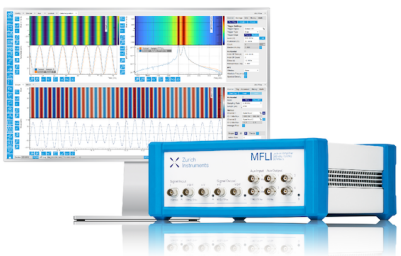
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TABLE II. Correction of electron diffraction and spectroscopic bond lengths to r_e.

	C—H in CH_4	C—D in CD_4
Electron diffraction		
r_g (exp)[a]	1.107_0	1.102_3
$r_g - r_e$ (calc)[b]	0.022_1	0.016_5
r_e	1.084_9	1.085_8
Spectroscopy		
r_0 (exp)	1.094_1[c]	1.092_2[d]
$r_0 - r_e$ (calc)[b]	0.009_1	0.006_8
r_e	1.085_0	1.085_4

[a] This investigation.
[b] Footnote reference 5.
[c] B. P. Stoicheff (private communication); K. T. Hecht (private communication).
[d] G. G. Sheppard and H. L. Welsh, J. Mol. Spectroscopy **1**, 277 (1957).

a perturbed normal coordinate treatment. The results are given in Table II.

No reliable derivation of r_e from the spectroscopically measured r_0 is available for comparison. Therefore, we have used the potential constants derived before to deduce the equilibrium distance, using the methods of Wilson and Howard,[5] and of Hecht.[6] The results, shown in Table II, indicate that the uncorrected spectroscopic and diffraction parameters differ appreciably from each other and from r_e. The excellent agreement between the equilibrium distances derived by the two methods lends weight to the present interpretation of the absolute significance of the parameters of each.

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† On leave from Department of Chemistry, Faculty of Science, University of Tokyo, Japan.

[1] R. A. Bonham and L. S. Bartell, J. Chem. Phys. **31**, 702 (1959).
[2] V. Schomaker and R. Glauber, Phys. Rev. **89**, 667 (1953); J. A. Ibers and J. A. Hoerni, Acta Cryst. **7**, 405 (1954).
[3] L. S. Bartell, J. Chem. Phys. **23**, 1219 (1955).
[4] K. Kuchitsu and L. S. Bartell, Abstract R7, Symposium on Molecular Structure and Spectroscopy, Columbus, Ohio, 1960.
[5] E. B. Wilson and J. B. Howard, J. Chem. Phys. **4**, 260 (1936).
[6] K. T. Hecht (private communication).

Dipole Moment of Lithium Hydride

LENNARD WHARTON,* L. PETER GOLD,† AND WILLIAM KLEMPERER‡

Department of Chemistry, Harvard University, Cambridge 38, Massachusetts

(Received August 8, 1960)

WE have measured the electric dipole moment of lithium hydride by the molecular beam electric resonance method.[1] Our apparatus utilizes electric quadrupole A and B fields for state selection and space focusing of the beam.[2] The uniform electric C field transition region is 15 cm long. The transition observed is the rotational angular-momentum reorientation $J=1, m_J=0 \rightarrow J=1, |m_J|=1$. For Li^7H in the ground-vibrational-state this transition was observed at about 3.2 Mc in an electric field of 730 v/cm, 22.7 Mc at 1920 v/cm, and 63.6 Mc at 3280 v/cm. The lines arising from the hyperfine structure of this transition extend over a range of 240 kc and the minimum width of these lines is 7 kc at optimum rf power. The hyperfine line pattern is identical at the three observed frequencies, with each line showing the expected strong field behavior

$$\nu_i(E) = \nu_{\mathrm{STARK}}(E) + \alpha_i,$$

where $\nu_{\mathrm{STARK}}(E)$ is the unperturbed Stark frequency and α_i is the voltage-independent displacement of a line from this frequency.

The electric dipole moments of Li^7H and Li^6H are[3]

Li^7H	$V=0$	$\mu=5.882\pm0.003D$
	$V=1$	$\mu=5.990\pm0.003D$
Li^6H	$V=0$	$\mu=5.884\pm0.003D$.

This research has been aided by support from the Atomic Energy Commission and the Research Corporation.

* Junior Fellow, Society of Fellows.
† National Science Foundation Predoctoral Fellow.
‡ Alfred P. Sloan Fellow.

[1] N. F. Ramsey, *Molecular Beams* (Oxford University Press, New York, 1956); P. Kusch and V. Hughes, *Encyclopedia of Physics*, (Springer-Verlag, Berlin, Vienna, 1959) Vol. 37.
[2] G. Bennewitz, W. Paul, and C. Schlier, Z. Physik **141**, 6 (1955); J. P. Gordon, H. J. Zeiger, and C. H. Townes, Phys. Rev. **99**, 1264 (1955).
[3] The rotational constants of Li^7H, $V=0$ and $V=1$, are the effective values of F. H. Crawford and T. Jorgensen, Phys. Rev. **47**, 932 (1935), while that of Li^6H is calculated from these constants.

Comments and Errata

Energy Levels of a Slightly Asymmetric Top

JOHN E. BEAM AND H. L. DAVIS

Physical Sciences Research Department, Sandia Corporation, Albuquerque, New Mexico

(Received July 25, 1960)

IT is the purpose of this comment to correct a table given by Norris and Laurie[1] for certain coefficients C_6 in the expansion

$$w = K^2 + \sum_{n=1}^{\infty} C_n b_p^n, \qquad (1)$$

for the reduced energy levels of a slightly asymmetric prolate rotor.[2] The present authors have been recently

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TABLE I. Sixth-order coefficients in the expansion of energy levels of the asymmetric rotor.

J	$K=0$	$K=2$	$\times 10^n$
5	−0.1298858	0.1298880	06
6	−0.1010301	0.1010369	07
7	−0.5782829	0.5783392	07
8	−0.2642748	0.2643054	08
9	−0.1016009	0.1016139	09
10	−0.3405496	0.3405958	09
11	−0.1020965	0.1021109	10
12	−0.2790172	0.2790576	10
13	−0.7052294	0.7053337	10
14	−0.1667336	0.1667587	11
15	−0.3720686	0.3721251	11
16	−0.7894026	0.7895234	11
17	−0.1601961	0.1602208	12
18	−0.3124980	0.3125464	12
19	−0.5884408	0.5885323	12
20	−0.1073401	0.1073569	13
21	−0.1902595	0.1902893	13
22	−0.3285454	0.3285970	13
23	−0.5539867	0.5540739	13
24	−0.9139532	0.9140972	13
25	−0.1477850	0.1478083	14
26	−0.2345803	0.2346174	14
27	−0.3660206	0.3660785	14
28	−0.5620897	0.5621789	14
29	−0.8504919	0.8506268	14
30	−0.1269197	0.1269399	15
31	−0.1869696	0.1869993	15
32	−0.2721120	0.2721552	15
33	−0.3915424	0.3916047	15
34	−0.5573859	0.5574747	15
35	−0.7854990	0.7856242	15
36	−0.1096458	0.1096633	16
37	−0.1516768	0.1517010	16
38	−0.2080339	0.2080671	16
39	−0.2830265	0.2830717	16
40	−0.3820980	0.3821590	16

engaged in obtaining general formulas for C_6 and C_7 in the expansion (1) using a form of perturbation theory based on the adiabatic theorem.[3] When comparing our work with the table given by Norris and Laurie, discrepancies were found to exist; furthermore, our work was also found to disagree with that of Polo.[4] It was then decided to explicitly check some of the entries of the table of footnote reference 1 using Eqs. (3) of Nielsen[5]; e.g., for the level $5_{2,3}$ we found that $C_6=[(C_2)^3-16C_2C_4-528C_4]/48$. Using known values[6] of C_2 and C_4, we obtain $C_6=0.1298880\times10^6$ which agrees with our general formula for C_6 for the $J_{2,J-2}$ level; however, both footnote references 1 and 4 give $C_6=0.1300071\times10^6$ for the $5_{2,3}$ level. Hence, one must conclude that the work of both footnote references 1 and 4 is in error; in fact, not one entry of the table of footnote reference 1 is found to agree with our general formulas. However, we have found minor errors in the work of Polo, which, when accounted for, make his and our results agree. We have calculated the values of C_6 for the levels $J_{0,J}$ and $J_{2,J-2}$ for $J=5$ through 40 using our general formulas; the results are given in Table I under the headings $K=0$ and $K=2$. The coefficients appearing in Table I are to be multiplied by the power of 10 given in the last column. Details of the method used to obtain general formulas for C_6 and C_7 and the formulas themselves will be submitted for publication in the near future.

[1] J. A. Norris and V. W. Laurie, J. Chem. Phys. **32,** 1591 (1960).
[2] See footnote reference 6 for the relation between the reduced energy levels and the actual energy levels of an asymmetric rotor.
[3] M. Gell-Mann and F. Low, Phys. Rev. **84,** 350 (1951).
[4] S. R. Polo, Can. J. Phys. **35,** 880 (1957).
[5] H. H. Nielsen, Phys. Rev. **38,** 1432 (1931). Corrections to Eqs. (3) of Nielsen's paper are found in H. M. Randall, D. M. Dennison, N. Ginsburg, and L. R. Weber, Phys. Rev. **52,** 160 (1937).
[6] C. H. Townes and A. L. Schawlow, *Microwave Spectroscopy* (McGraw-Hill Book Company, Inc., New York, 1955).

Erratum: Energy Levels of a Slightly Asymmetric Top

[J. Chem. Phys. **32,** 1591 (1960)]

JAMES A. NORRIS AND VICTOR W. LAURIE

Department of Chemistry, University of California, Berkeley 4, California

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IT has been pointed out to us by Dr. J. E. Beam and Dr. H. L. Davis of Sandia Corporation that the formulas of Polo[1] which were the basis for the calculation of our tables contain minor errors, and that therefore the tables are not quite correct. The absolute magnitude of all entries is too large. For $J>25$, the largest error is $<0.01\%$. For $K=0$, $J=5$–10, however, errors of $\sim1\%$ are present. Corrected tables will be published by Beam and Davis elsewhere in this issue.[2]

[1] S. R. Polo, Can. J. Phys. **35,** 880 (1957).
[2] J. E. Beam and H. L. Davis, J. Chem. Phys. **33,** 1256 (1960).

Analysis of High-Resolution NMR Spectra by Iterative Methods

RAGNAR A. HOFFMAN

Department of Physics, University of Uppsala, Uppsala, Sweden

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IN a recent paper, Reilly and Swalen[1] have advanced some iterative procedures for analyzing NMR spectra. The schemes denoted[2] ABK and $AA'K$ will be found useful for analyses of certain types of three-spin systems. The iterative procedure derived for ABC-cases is applicable also for systems of more than three spins, and the purpose of the present note is to compare this method (I) with a closely related method (II), which is based[3] on perturbation theory.

A problem generally encountered in spectroscopy is to determine from a number of observed transition energies and intensities a set of parameters $(a, b, c, \cdots)$ of the Hamiltonian H $(a, b, c, \cdots)$, which describes the

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